Respiration, exercise & metabolism

Transfer energy

1 Aerobic respiration

Glucose + oxygen → CO_2 + water. Mostly in mitochondria; complete oxidation.

Exothermic, continuous in living cells; energy for movement, keeping warm and building molecules.

Respiration is not breathing; energy is transferred, not created.

2 Anaerobic & fermentation

Muscles: **glucose → lactic acid.** Yeast/plants: **glucose → ethanol + CO_2.** No oxygen required.

Incomplete breakdown transfers much less energy per glucose than aerobic respiration.

Yeast fermentation: CO_2 raises bread dough; ethanol makes alcoholic drinks.

Respond to demand

3 During exercise

Heart rate and breathing rate/volume increase, delivering more oxygenated blood to muscles.

If oxygen supply cannot meet demand, anaerobic respiration increases → lactic acid builds up.

Prolonged vigorous activity → fatigue and less efficient contraction.

4 Recovery & investigation

Oxygen debt: extra oxygen needed after exercise to deal with accumulated lactic acid. Blood carries it to liver → glucose.

Compare resting/post-exercise pulse; standardise time/pace, allow recovery, repeat and use safe activities.

15-second pulse count × 4 = beats/min.

Build & break molecules

5 Building molecules

Metabolism = all reactions in cells/body. Enzymes control reactions; respiration supplies energy.

Glucose → starch (plants), glycogen (animals), cellulose (plant walls).

1 glycerol + 3 fatty acids → lipid.

6 Protein & waste

Plants: glucose + nitrate-derived nitrogen → amino acids → proteins.

Excess proteins/amino acids are broken down, producing **urea for excretion.**

Digestion supplies small molecules; cells rebuild their own materials. Respiration is one part of metabolism.